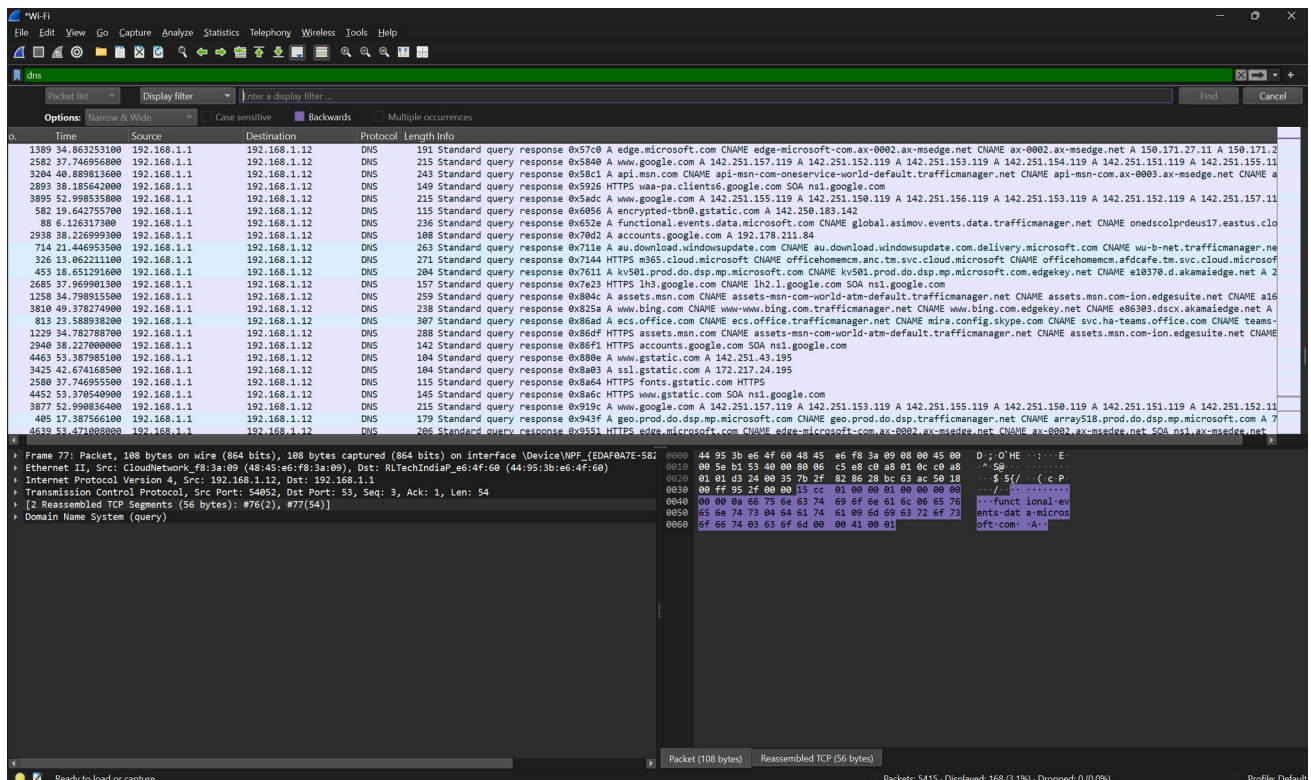


CO5 -ASSESSMENT TOOL 1 - Constraint-Based

Problem Solving

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COMPUTER NETWORKS

TASK 1 : DNS Resolution - Google:



- Packet 2582 is the DNS response that resolves www.google.com.

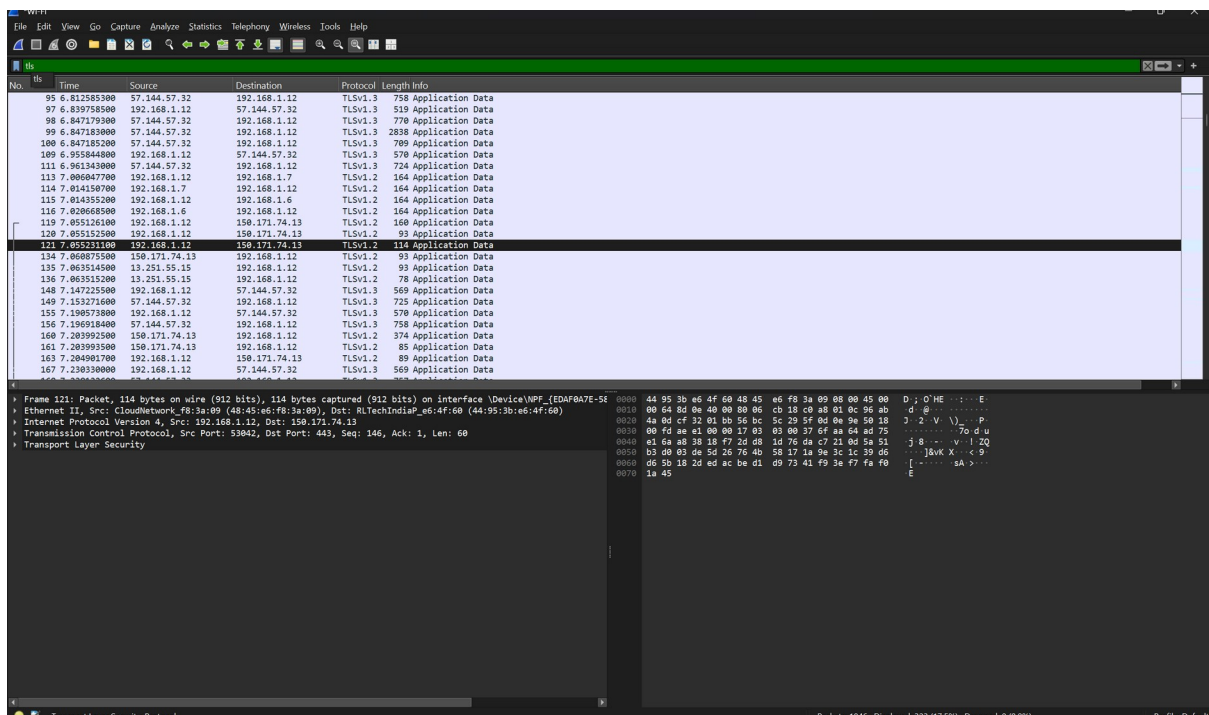
Packet Details:

- Packet Number: 2582
- Transaction ID: 0x5840
- Info Column: Standard query response 0x5840 A www.google.com A 142.251.157.119...

Task Form Completion

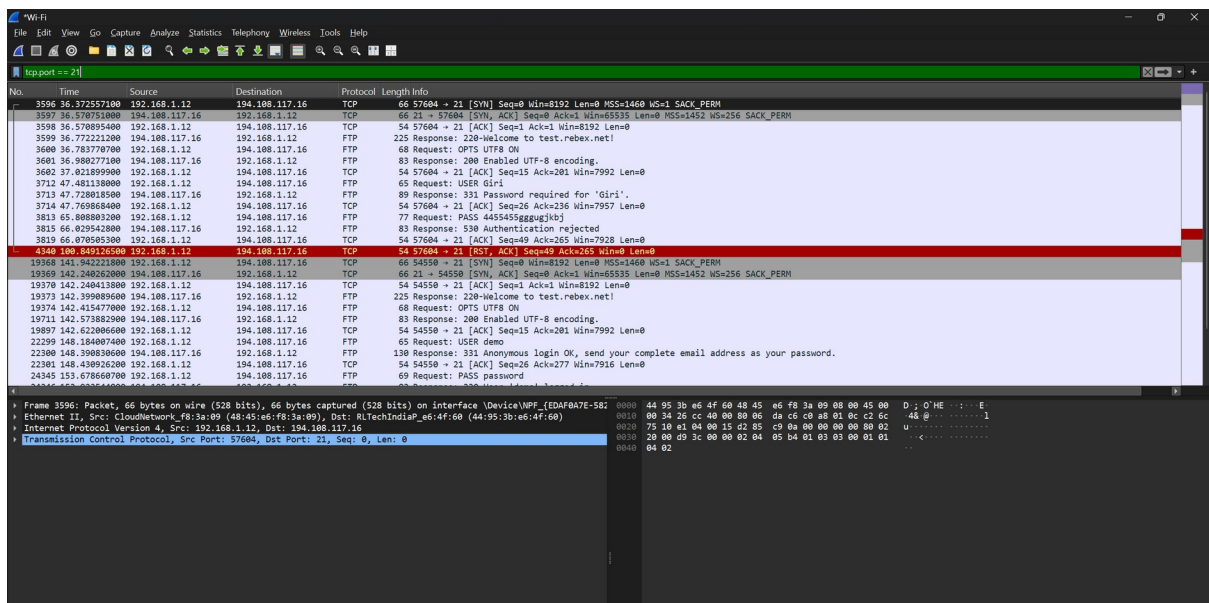
- Digital response: 2582
- Wireshark filter: dns (or dns.flags.response == 1 && dns.qry.name == "www.google.com")

TASK 2 - HTTPS Web Access - Wikipedia:



- Correct answer: TCP
- Wireshark filter: `tls (or tls.handshake.type == 1)`

TASK 3 - FTP Traffic Challenge - Real FTP Test Service:

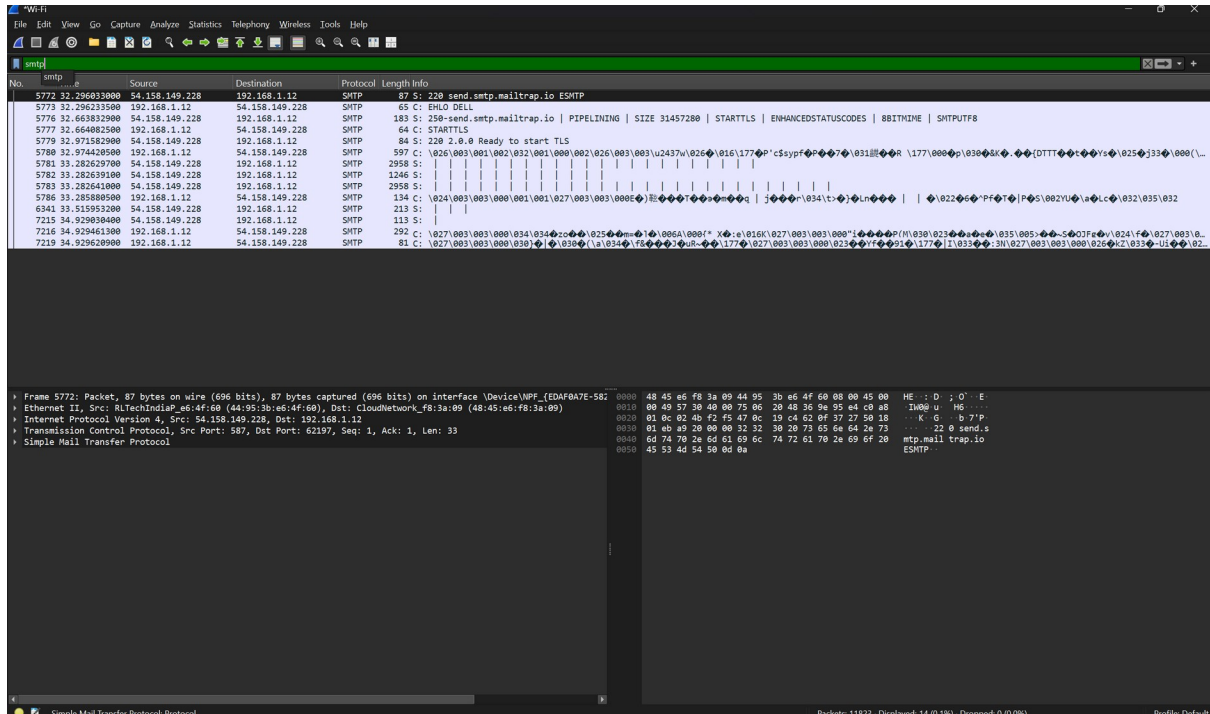


FTP Control Connection :

- Correct answer: Port 21

- Wireshark filter: tcp.port == 21.

TASK 4 -SMTP Traffic Identification - Mailtrap Sandbox:



```
PS C:\Users\Giridhar> $secpasswd = ConvertTo-SecureString "c3a8d94873238b92ce38785bec43ab54" -AsPlainText -Force
PS C:\Users\Giridhar> $cred = New-Object System.Management.Automation.PSCredential("api", $secpasswd)
PS C:\Users\Giridhar> Send-MailMessage -From "test@demomailtrap.co" -To "g6442117@gmail.com" -Subject "SMTP Test" -Body
"Testing" -SmtpServer "live.smtp.mailtrap.io" -Port 587 -UseSsl -Credential $cred
```

- Protocol Name: Simple Mail Transfer Protocol (SMTP)
- OSI Layer: Application Layer (Layer 7)
- Observed Port: 587 (Submission / STARTTLS)
- **Correct Response = SMTP.**

```
=====
=====
```